

Eranga Kavisanka (He/Him)

Software Engineering Undergraduate | Freelancer



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OVERVIEW

Software Engineering undergraduate with hands-on experience developing production-ready web and cloud-native applications. Strong interest in software engineering, distributed systems, cloud-native technologies, DevOps, and Site Reliability Engineering (SRE), including industry-recognized certifications in **AWS Certified Cloud Practitioner (AWS CCP)** and **Kubernetes and Cloud Native Associate (KCNA)**. **Batch Top**, **published researcher**, **open-source contributor**, and **curious learner** passionate about solving complex engineering problems through scalable software.

EDUCATION

B.Comp. (Hons) in Software Engineering (UG), 2023 – present
University of Sri Jayewardenepura

GPA: **3.89/4**, **Batch Top** in Software Engineering Degree Program

Expertise: *Cloud Computing, Object-Oriented programming, Software Engineering, Programming Languages, Data Structures and Algorithms, Artificial Intelligence, Machine Learning, Deep Learning, Computer Networks*

GCE Advanced Level Examination (Physical Science), Pinnawala Central College - Kegalle 2021

OPEN SOURCE CONTRIBUTIONS

WSO2 Open Source Contributor – Contributed to the WSO2 Docs Integrator project with a merged pull request improving technical documentation [[CONTRIBUTION](#)]

MAYHEM Kubernetes Incident Simulator – Contributed to testing and validation, providing technical feedback; received a recommendation from the project's DevOps & SRE Tech Lead. [[RECOMMENDATION](#)]

PROFESSIONAL EXPERIENCE

Co-Founder & Full Stack Software Developer, InventX Solutions 2024 – Present

Co-founded a software startup delivering production-ready web and mobile applications for clients, participating in requirements analysis, system design, development, deployment, and maintenance.

- *Sekin Express Delivery Platform* - Delivered a production-ready logistics management platform for a real-world client, enabling secure multi-role operations, parcel tracking, and centralized delivery management. [[Live](#)]
- *QS Monach Corporate Website* — Responsive corporate website with a modern user interface and performance optimization. [[Live](#)]
- *SkillFinder Mobile Application* — Flutter mobile application connecting users with skilled professionals with an integrated administrative portal. [[GitHub](#)]

Banking Intern, People's Bank October 2022 – June 2023

Supported daily branch operations, customer service, and administrative banking processes.

RESEARCH PUBLICATIONS

A Review and Comparative Analysis of RF-Based Offline Communication Systems and IoT Integration Approaches for Traveller and Hiker Safety

Jayewardenepura Computing Symposium (JCoS) 2026

Reviewed and compared existing RF-based communication technologies, mesh networking protocols, and IoT architectures, identifying challenges and future directions for resilient off-grid emergency communication systems.

CERTIFICATIONS

AWS Certified Cloud Practitioner (AWS CCP)

Amazon Web Services

Kubernetes and Cloud Native Associate (KCNA)

Linux Foundation

Aviatrix Certified Multicloud Network Associate (ACE)

Aviatrix

Linux Foundation Training- LFS101, LFS158, LFS162, LFS167, LFS169

Linux Foundation

Full Stack Web Development

Skyrek

SELECTED PROJECTS AND EXPERIENCES

Hybrid RF-Based Off-Grid Mesh Communication Network with Opportunistic Cloud Synchronization (Ongoing Final Year Project)

Tech Stack: ESP32-C3, IoT, AWS IoT Core, MQTT, Grafana, Flutter, MySQL

Developing a hybrid RF mesh communication system enabling offline emergency messaging, opportunistic cloud synchronization, and real-time IoT monitoring. Integrating 6+ cloud and edge components including AWS IoT Core, MQTT, local persistence, and Grafana dashboards while strengthening expertise in distributed IoT systems, embedded communication, and cloud-connected edge architectures. [\[SOURCE\]](#) [\[ARCHITECTURE\]](#)

Cloud-Native GitOps Platform on Amazon EKS

Tech Stack: Amazon EKS, Terraform, GitHub Actions, Argo CD, Helm, Docker, Amazon ECR, SonarQube

Engineered a GitOps platform spanning 3 repositories (Application, GitOps, Infrastructure) with automated infrastructure provisioning, CI/CD, quality gates, container delivery, and continuous reconciliation. Strengthened expertise in GitOps workflows, Infrastructure as Code, Kubernetes operations, and production-grade cloud-native deployment automation. [\[SOURCE\]](#) [\[APP\]](#) [\[GITOPS\]](#) [\[IaC\]](#)

Distributed Multi-Tier Java Application on Kubernetes

Tech Stack: Kubernetes, kOps, AWS EC2, Docker, NGINX Ingress, Route53, Amazon EBS CSI, MySQL, RabbitMQ, Memcached

Deployed a production-style 5-tier Java application using 8+ interconnected services with persistent storage, ingress, Route53 DNS, and service discovery. Enhanced expertise in Kubernetes networking, storage orchestration, workload scheduling, container networking, and troubleshooting distributed cloud-native applications. [\[SOURCE\]](#) [\[ARCHITECTURE\]](#)

Crystal Beauty – Production-Ready Full-Stack MERN E-Commerce Platform

Tech Stack: React, Node.js, Express.js, MongoDB, Tailwind CSS, Google OAuth, Supabase, Netlify, Render

Designed and developed a full-stack e-commerce platform featuring secure authentication, role-based access control, product management, shopping cart, order processing, customer reviews, and Google OAuth integration. Deployed the application using modern cloud platforms with automated deployment pipelines and environment-based configuration. [\[FRONTEND\]](#) [\[BACKEND\]](#) [\[DEMO\]](#)

Production-Grade Cloud Observability Platform on AWS

Tech Stack: Prometheus, Grafana, Loki, Alloy, Flask, AWS, Slack

Implemented centralized metrics, logging, and alerting across 4 AWS EC2 instances using Prometheus, Grafana, Loki, and Alloy. Built production dashboards, PromQL-based monitoring, and Slack notifications while developing practical SRE expertise in observability, incident detection, log analysis, and root-cause investigation. [\[SOURCE\]](#) [\[ARCHITECTURE\]](#) [\[DEMO\]](#)

Serverless AWS Security Compliance Automation Platform

Tech Stack: AWS Lambda, Python (boto3), EventBridge, SNS, S3

Developed a serverless security auditing solution scanning AWS Security Groups across multiple AWS regions, generating compliance reports, storing audit logs in Amazon S3, and delivering automated SNS notifications. Strengthened expertise in serverless computing, event-driven architectures, AWS Lambda, Python automation, and cloud security. [\[SOURCE\]](#) [\[ARCHITECTURE\]](#)

Jenkins-Driven CI/CD Pipeline for Containerized Applications on AWS ECS

Tech Stack: Jenkins, Docker, Maven, SonarQube, Amazon ECS, Amazon ECR

Built an end-to-end CI/CD pipeline integrating 6+ automation stages including Maven builds, SonarQube analysis, Docker image creation, Amazon ECR publishing, and ECS deployments. Enhanced expertise in enterprise CI/CD, container delivery, code quality automation, and release engineering. [\[SOURCE\]](#) [\[ARCHITECTURE\]](#) [\[DEMO\]](#)

Multi-Node Infrastructure Automation using Ansible on AWS

Tech Stack: Ansible, AWS EC2, Linux, SSH, Apache HTTP Server

Automated provisioning and configuration across multiple EC2 instances using Ansible inventory groups, ad-hoc commands, and agentless SSH. Improved deployment consistency while gaining practical experience in configuration management, Linux administration, infrastructure automation, and scalable server provisioning. [\[SOURCE\]](#)

Enterprise DevSecOps CI/CD Pipeline with Automated Security & Quality Gates

Tech Stack: GitHub Actions, Maven, Docker, Trivy, Amazon ECR, Java

Built a production-ready DevSecOps CI/CD pipeline for the vProfile application, automating software delivery with integrated code quality analysis, vulnerability scanning, container image security, and artifact publishing using GitHub Actions, Maven, Trivy, Docker, and Amazon ECR. [\[SOURCE\]](#) [\[ARCHITECTURE\]](#)

Full-Stack E-Channeling Management System (University Group Project)

Tech Stack: Laravel, PHP, MySQL, Stripe, JavaScript

Developed the backend architecture for a production-style healthcare appointment management system supporting 5+ user roles, secure authentication, online appointment scheduling, Stripe payment integration, PDF receipt generation, and over 240+ automated functional and security test cases. Collaborated within a multidisciplinary team to deliver a scalable and secure full-stack web application. [\[SOURCE\]](#)

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, Python, PHP, C, Bash

Web & Backend Development: React, Node.js, Express.js, Laravel, REST APIs, Flutter, n8n

Databases: MySQL, MongoDB

Cloud Platforms: Amazon Web Services (AWS)

Containers & Cloud-Native: Docker, Docker Compose, Kubernetes, Amazon EKS, Helm, kOps

CI/CD, GitOps & DevSecOps: GitHub Actions, GitLab CI/CD, Jenkins, Argo CD, SonarQube, Trivy

Infrastructure as Code: Terraform, Ansible, Vagrant

Observability: Prometheus, Grafana, Loki, Alloy, Amazon CloudWatch

AWS Services: EC2, EKS, ECR, ECS, IAM, VPC, Route 53, S3, Lambda, SNS, EventBridge, Elastic Beanstalk, Amazon MQ

Linux & Networking: Ubuntu, CentOS, Amazon Linux, NGINX, DNS, Load Balancing, MQTT

Version Control: Git, GitHub, GitLab, Bitbucket

OTHER SKILLS

Problem Solving	Strong analytical and problem-solving skills with attention to detail
Curiosity	Passionate about exploring emerging technologies and continuously expanding technical knowledge
Teamwork	Effective collaboration in cross-functional and team environments
Communication	Professional communication in English and Sinhala
Time Management	Ability to work independently and manage multiple tasks
Teaching & Mentoring	Experience mentoring and teaching technical concepts

LEADERSHIP, COMMUNITY & TECHNICAL CONTRIBUTIONS

Technical Lead, AWS Student Builder Group, University of Sri Jayewardenepura	2026 – Present
Head of Teaching Panel, FusionX 1.0 AI Bootcamp (SLAAI)	2025
Top 30 Finalist, HaXtreme 4.0 National Coding Hackathon	2025
Batch Representative, Faculty of Computing, University of Sri Jayewardenepura	2025 – Present
IEEE Student Branch Member, University of Sri Jayewardenepura	2024 – 2025
Author, Kubernetes Hidden Gems (LinkedIn Technical Newsletter, 1K+ Subscribers)	2026 – Present
Technical Blogger, Medium	2025 – Present

BLOGS

Getting Started with AWS EKS: A Practical Guide to Managed Kubernetes on AWS
An introductory guide to deploying and managing Kubernetes clusters using Amazon EKS. [\[Read more\]](#)

Model Context Protocol (MCP) Introduction
An introduction to MCP and how AI models securely interact with external tools and services. [\[Read more\]](#)

Kubernetes Hidden Gems #2 — Startup Probes
Exploring Startup Probes for improving application startup reliability in Kubernetes. [\[Read more\]](#)

Kubernetes Hidden Gems #6 — Pod Disruption Budgets (PDBs)
Understanding Pod Disruption Budgets for maintaining application availability during disruptions. [\[Read more\]](#)

[More articles on Medium](#)

[More on Kubernetes Hidden Gems Newsletter](#)

REFEREES

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